

Cambridge IGCSE™

CHEMISTRY**0620/43**

Paper 4 Theory (Extended)

May/June 2024

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre–U components, and some Cambridge O Level components.

This document consists of **12** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science–Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non–contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1(a)	Haber (process)	1
1(b)	electrolysis	1
1(c)	filtration	1
1(d)	catalytic addition	1
1(e)	chromatography	1
1(f)	fractional distillation	1
1(g)	titration	1

Question	Answer	Marks
2	M1 Cu: 29 and 34 M2 Cl^- : 17 and 18 M3 $^{64}_{30}$ to left of symbol M4 Zn M5 2^+	5

Question	Answer	Marks
3(a)(i)	mobile electrons	1
3(a)(ii)	M1 double bond of two dots and two crosses M2 4 non-bonding electrons on each oxygen atom	2
3(a)(iii)	M1 liquid M2 melting point is below -195°C and boiling point is above -195°C or -195°C is higher than -199°C / melting point and lower than -191°C / boiling point or -195°C is in between melting point / -199°C and boiling point / -191°C	2
3(a)(iv)	M1 graphite has a giant covalent structure M2 attraction between molecules in carbon monoxide M3 weaker attraction (between particles) in carbon monoxide ORA	3
3(b)	$2\text{K} + \text{Cl}_2 \rightarrow 2\text{KCl}$ M1 KCl as only product M2 correct equation	2
3(c)(i)	M1 breakdown by (the passage of) electricity M2 of an ionic compound in molten or aqueous (state)	2

Question	Answer	Marks
3(c)(ii)	$4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ M1 any negatively charged OH species losing electrons M2 correct ionic half-equation	2

Question	Answer	Marks				
4(a)	M1 (rates are) equal M2 (concentrations are) no longer changing	2				
4(b)	more molecules of nitrogen dioxide per unit volume OR same number of nitrogen dioxide molecules in a smaller volume	1				
4(c)	(the forward reaction is) endothermic	1				
4(d)(i)	<table border="1"><tr><td>M1 increases</td><td>M3 no change</td></tr><tr><td>M2 increases</td><td>M4 increases</td></tr></table>	M1 increases	M3 no change	M2 increases	M4 increases	4
M1 increases	M3 no change					
M2 increases	M4 increases					
4(d)(ii)	M1 (the oxidation number of sulfur in S) = zero M2 (the oxidation number of sulfur in SO ₃) = +6	2				

Question	Answer	Marks
5(a)(i)	M1 named soluble barium salt e.g. barium chloride / barium nitrate M2 named soluble sulfate salt e.g. sodium sulfate / potassium sulfate / ammonium sulfate / sulfuric acid	2
5(a)(ii)	M1 filter M2 wash (residue) with water and dry (residue) between filter papers / in a warm place	2
5(a)(iii)	$\text{Ba}^{2+} (\text{aq}) + \text{SO}_4^{2-} (\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ M1 BaSO_4 as only product M2 $\text{Ba}^{2+} + \text{SO}_4^{2-}$ as only reactants M3 state symbols	3
5(b)	M1 sulfuric (acid) M2 copper(II) carbonate / copper(II) oxide / copper(II) hydroxide	2
5(c)(i)	glowing splint and relights	1
5(c)(ii)	M1 $2\text{CuO} + 4\text{NO}_2$ M2 $6\text{H}_2\text{O}$	2
5(d)(i)	(To ensure that) all the water is given off	1

Question	Answer	Marks
5(d)(ii)	M1 mol ZnSO_4 ($0.322 / 161 =$) 0.002(00) M2 mass H_2O given off ($0.574 - 0.322 =$) 0.252 M3 mol H_2O given off ($0.252 \div 18 =$) 0.014(0) M4 value of x ($0.014 \div 0.002 =$) 7	4

Question	Answer	Marks
6(a)(i)	hematite	1
6(a)(ii)	air	1
6(a)(iii)	carbon monoxide	1
6(a)(iv)	M1 limestone thermally decomposes (into calcium oxide and carbon dioxide) M2 calcium oxide reacts with silicon(IV) oxide M3 producing slag	3
6(b)(i)	have low densities or have low melting point	1
6(b)(ii)	act as catalysts or form coloured compounds	1

Question	Answer	Marks
6(b)(iii)	form basic oxides or form positive ions	1
6(b)(iv)	form acidic oxides or form negative ions	1
6(c)(i)	M1 magnesium loses electrons more readily than / in preference to / instead of iron M2 magnesium is more reactive than iron ORA	2
6(c)(ii)	any metal below iron in the reactivity series e.g. copper / silver / gold	1

Question	Answer	Marks
7(a)(i)	M1 C 83.33 / 12: H 16.67 / 1 or 6.94:16.67 M2 C 6.94 / 6.94: H 16.67 / 6.94 or 1:2.4 or 5:12 M3 C₅H₁₂	3
7(a)(ii)	M1 M_r of C₂H₅ = 29 M2 58 / 29 = 2 and C₄H₁₀	2
7(b)(i)	C=C	1
7(b)(ii)	M1 $\begin{array}{c} \quad \\ \text{C} = \text{C} \\ \quad \end{array}$ any C=C with both carbons having a valency of 4 M2 displayed formula of but-1-ene M3 but-1-ene	3

Question	Answer	Marks
7(c)	M1 $2\text{C}_2\text{H}_4$ M2 C_8H_{18}	2
7(d)(i)	$\text{C}_n\text{H}_{2n} + \text{OH}$	1
7(d)(ii)	M1 displayed formula of propan-1-ol or propan-2-ol M2 propan-1-ol or propan-2-ol	2